

Overnight search experiments

IQ-TREE search-strategy tests run on 10–11 September 2026 · laptop (16 logical cores, 7 GB RAM) · local build of IQ-TREE 3.1.3 with the search-experiment additions

What ran. Two scripts, chained so they never shared the machine's RAM:

Run	Question	Data	When	Runs
1 · misof test	Does SPR weightprune (SPR kick + SPR refinement) beat stock NNI on a large empirical alignment?	misof DNA v1 (144 taxa × 100,575 sites), seed 777	10 Sep 18:08–23:56	2
2 · slack sweep	Does bounding the kick's damage ("slack"), or replacing the kick with stochastic acceptance, beat stock IQ-TREE?	covid_500 (500 taxa × 29,903 sites), seeds 777, 1234	6 runs 10 Sep 12:45–16:39; 20 runs 11 Sep 00:23–06:08	26

Findings.

- **misof: same tree, 21× the CPU.** SPR weightprune finished on the identical topology as stock NNI (RF = 0; both 10 of 282 splits from the true tree). Neither search loop improved on the best starting tree; the SPR arm spent 37,810 CPU-seconds of tree search to the NNI arm's 1,785.
- **Slack helps on covid_500, most of all when annealed.** Bounding each kick move to 1 or 5 logL beat stock IQ-TREE on both seeds. The two strongest arms were the annealed SPR-kick bound of 1 (+28.9 logL mean, nearly identical on both seeds) and the annealed NNI-kick bound of 5 (+16.9 mean, at the lowest cost of the winners).
- **Three things lost to stock:** a loose bound of 20, a bare SPR kick, and accept-dist (stochastic acceptance with no kick).
- **The slack gains are not yet proven at equal compute.** Every arm ran a fixed 20 iterations, and the winners used 6–15× the control's CPU. The annealed and accept-dist arms also ran with bugs that were fixed the next morning (see *Limitations*).

1 · misof: stock NNI vs SPR weightprune

Setup

Alignment	<code>SpeciesTreeGen/run_misof_empirical_dna/concatenated_alignment.fasta</code> — 144 taxa, 100,575 columns, 100,560 distinct patterns, 86,971 parsimony-informative sites; simulated, with a known true species tree
Shared settings	<code>-st DNA -m GTR+F -nt 2 --ninit 10 --ntop 5 -nstop 30 -seed 777</code>
Control	stock IQ-TREE (NNI kick, NNI refinement); no search flags
SPR arm	adds <code>--spr-perturb "radius 6 weightprune"</code> and <code>--spr-refine "radius 10 fast quiet fullreopt 100 20 weightprune"</code>
Stopping	each arm stops on IQ-TREE's own rule: 30 iterations without improvement

Tree-search CPU, SPR vs NNI

21×

37,810 s vs 1,785 s

RF distance between final trees

0

identical topology

Improvements after iteration 1

0

in either arm

Results

	Stock NNI	SPR weightprune
Final log-likelihood	−6,044,156.334	−6,044,156.287
Best tree during search (found at iteration 1)	−6,044,156.518	−6,044,156.518
Iterations	32	32
Tree-search CPU	1,785 s	37,810 s
Total CPU	2,049 s	38,038 s
Wall clock	21 min	5 h 29 min
Candidate moves evaluated	28,304	65,691
CPU per candidate	63 ms	576 ms
RF distance to true tree	10 / 282	10 / 282

What the numbers say

The search loop did no work in either arm. Both runs' best tree came out of the starting-tree phase (iteration 1), and 31 further iterations found nothing better. The 0.047 logL difference in the final scores is the final branch-length and model optimization settling slightly differently on the same topology. The 10 splits separating both trees from the true tree are inference error from the data, not search error.

The SPR arm could not recover from its own kicks. The tree each iteration handed back to IQ-TREE was only a little below the best tree for NNI, but tens of thousands of logL below it for SPR:

Distance below the best tree	Iteration 10	Iteration 20	Iteration 30
Stock NNI	8.3	18.6	12.2
SPR weightprune	24,445	37,265	25,995

The SPR kick makes as many moves as the NNI kick would (70 on 144 taxa), each up to radius 6. With 100k sites behind the likelihood, one badly placed subtree can cost thousands of logL, far more than the `fast` refinement repairs.

Where the 21× comes from. The SPR arm evaluated 2.3× as many candidate moves and each cost 9.1× as much. A side benchmark on a 4,000-site subset of the same taxa (1 thread, 20 iterations) split the per-run cost into the SPR algorithm itself (≈3.5× NNI), `fullreopt 100 20` (≈1.5×) and the safe numerical kernel that `fullreopt` switches on (≈1.1×; it costs NNI 1.39×, so it does not hit SPR disproportionately). That totals ≈5.9× at 4k sites against 21× at 100k: the extra cost appears only on the full-length alignment, and sits in the cost *per candidate*, not the number of candidates.

Consistent with the earlier test. An earlier comparison on all three misof replicates (v1–v3), using `--spr-continuous` with a step budget matched to NNI's CPU, also gave identical scores and RF = 0 on every replicate.

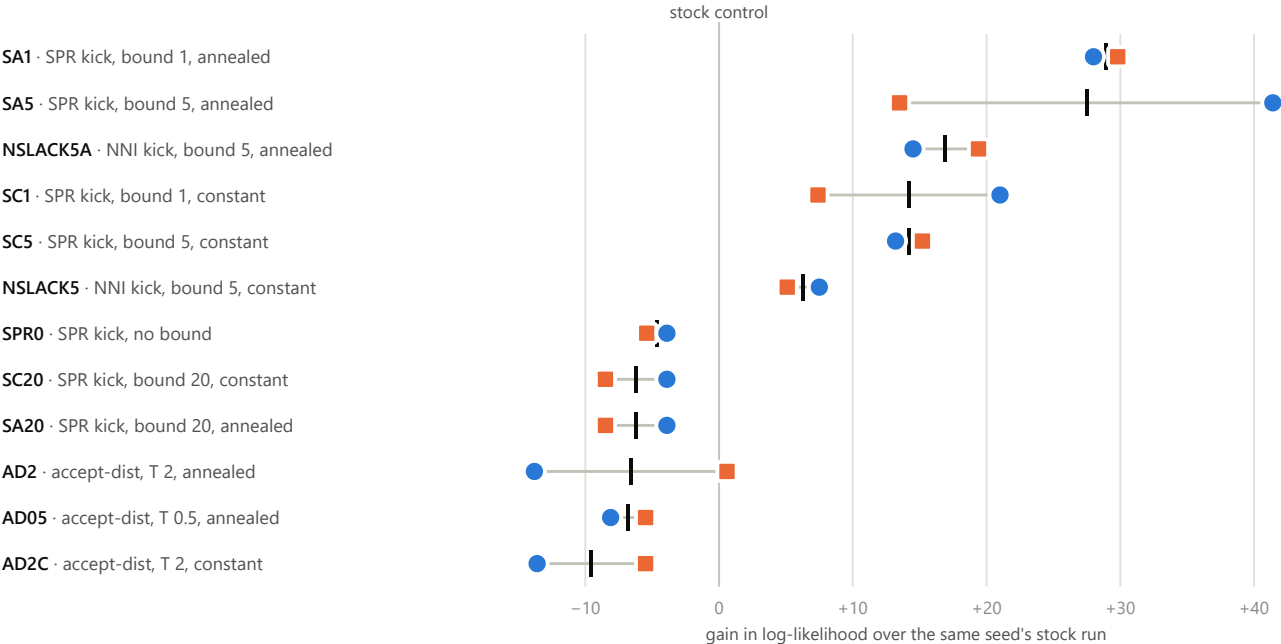
2 · covid_500 slack sweep

Setup

Alignment	covid_data/covid_500.fasta — 500 SARS-CoV-2 genomes, 29,903 columns, 8,179 distinct patterns, 1,214 parsimony-informative sites. The one alignment in this work where search choices have changed the result.		
Shared settings	-st DNA -m GTR+F -nt 2 --ninit 10 --ntop 5 -n 20, seeds 777 and 1234. Every arm runs exactly 20 iterations.		
Gain	an arm's final logL minus the stock control's, for the same seed . Positive = better than stock IQ-TREE.		
Arm	Kick	Bound on each kick move	Flags added to the shared settings
CTRL	NNI (stock)	none	—
SPR0	SPR	none	--spr-perturb "radius 6"
SC1 · SC5 · SC20	SPR	1 · 5 · 20 logL, constant	--spr-perturb "radius 6 slack D"
SA1 · SA5 · SA20	SPR	1 · 5 · 20 logL, annealed	--spr-perturb "radius 6 slack D anneal"
NSLACK5 · NSLACK5A	NNI (stock)	5 logL, constant · annealed	--perturb-slack 5 [anneal]
AD05 · AD2 · AD2C	none	— (stochastic acceptance in the NNI refinement instead)	--accept-dist "temp 0.5 anneal" · "temp 2 anneal" · "temp 2"

Gain over stock IQ-TREE, by arm and seed

Seed 777 Seed 1234 | Mean of both seeds

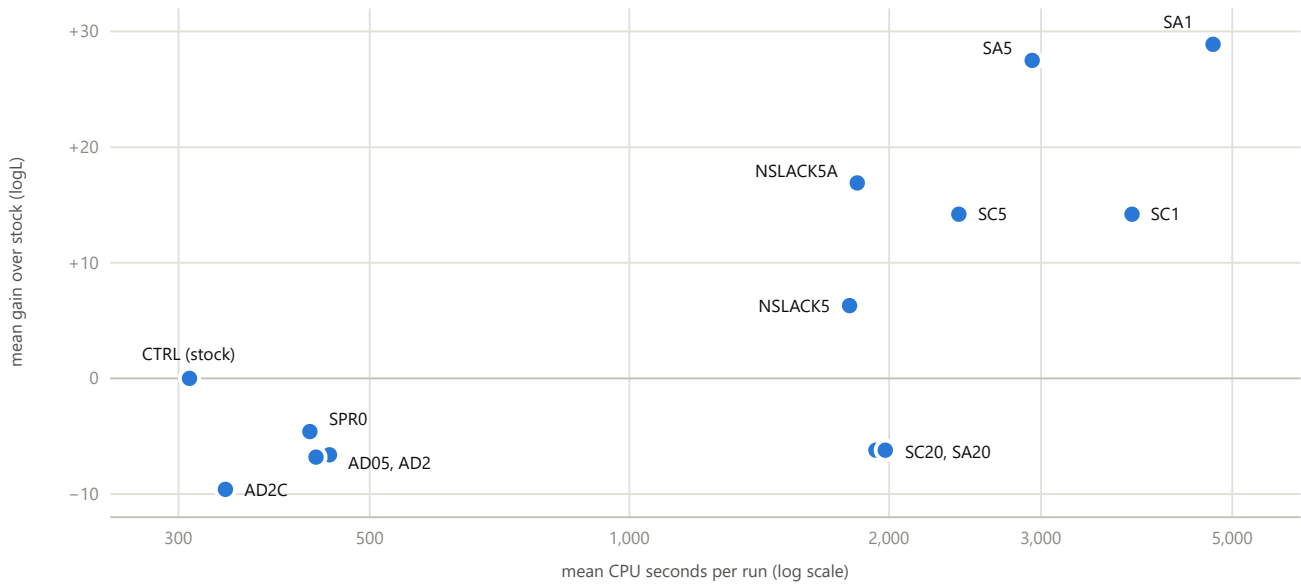


Rows are sorted by mean gain. The gap between a row's two markers is the seed-to-seed spread: small for SA1 and NSLACK5A, 28 logL for SA5. Exact values are in the table below.

Results

Arm	Gain, seed 777	Gain, seed 1234	Mean gain	Mean CPU	CPU vs control
SA1 · SPR kick, bound 1, annealed	+28.0	+29.8	+28.9	4,749 s	15.4×
SA5 · SPR kick, bound 5, annealed	+41.4	+13.5	+27.5	2,931 s	9.5×
NSLACK5A · NNI kick, bound 5, annealed	+14.5	+19.4	+16.9	1,837 s	5.9×
SC1 · SPR kick, bound 1, constant	+21.0	+7.4	+14.2	3,826 s	12.4×
SC5 · SPR kick, bound 5, constant	+13.2	+15.2	+14.2	2,409 s	7.8×
NSLACK5 · NNI kick, bound 5, constant	+7.5	+5.1	+6.3	1,800 s	5.8×
CTRL · stock IQ-TREE	—	—	0	309 s	1.0×
SPR0 · SPR kick, no bound	−3.9	−5.4	−4.6	426 s	1.4×
SC20 · SPR kick, bound 20, constant	−3.9	−8.5	−6.2	1,931 s	6.2×
SA20 · SPR kick, bound 20, annealed	−3.9	−8.5	−6.2	1,980 s	6.4×
AD2 · accept-dist, T 2, annealed	−13.8	+0.6	−6.6	449 s	1.5×
AD05 · accept-dist, T 0.5, annealed	−8.1	−5.5	−6.8	433 s	1.4×
AD2C · accept-dist, T 2, constant	−13.6	−5.5	−9.6	340 s	1.1×

Mean gain against mean CPU cost



Horizontal line = the stock control. Everything above it gained; everything to its right cost more. The arms that gained all sit 6–15× to the right: their extra CPU bought the gain, and no run tested whether stock IQ-TREE given the same CPU would do as well.

What held on both seeds

- **A bound of 1 or 5 helps; 20 does not.** All six arms with a bound of 1 or 5 beat stock on both seeds. A bound of 20 is too loose to refuse anything useful: on seed 777 SC20, SA20 and the bare SPR kick all finished on the identical tree.
- **Annealing usually helps.** Annealed beat constant in 5 of the 6 pairs where the bound mattered (bound 1: +7.0 and +22.4; SPR kick bound 5: +28.2 and −1.7; NNI kick bound 5: +7.0 and +14.3), roughly doubling the mean gain. With a bound of 20 both schedules gave the same result.
- **SA1 is the steadiest winner; NSLACK5A the cheapest.** SA1 gained +28.0 and +29.8. NSLACK5A gained +14.5 and +19.4 at about 1,840 CPU-seconds on both seeds. It is also the smallest change to stock IQ-TREE: its usual NNI kick plus `--perturb-slack 5 anneal`. SA5 averaged nearly as much as SA1 but swung by 28 logL between seeds.
- **A bare SPR kick is slightly worse than the stock NNI kick** (−3.9 and −5.4).
- **Accept-dist lost to stock** in 5 of 6 runs (the sixth gained just 0.6), while costing 1.1–1.5× the control's CPU. Replacing the kick with stochastic acceptance explored less than the kick it replaced. See the bug note below before drawing a final conclusion.

3 · Limitations

Both runs

- **Timings are CPU-seconds from a laptop.** Wall clock is unusable here; one earlier run in this project logged 9.5% CPU utilization because the machine slept through it. CPU-seconds are comparable within a run set, but inflate modestly when another job shares the machine. At least one of the six slack arms run on 10 September (SA1, seed 777) overlapped other tests, and the last minutes of the misof SPR run overlapped a small benchmark.
- **This build uses the SSE2 kernel, not AVX.** Absolute times are slower than an official IQ-TREE release on the same machine; only the ratios between arms carry over.
- **Few replicates, no significance testing.** The misof test has one run per arm and the slack sweep two per arm, so effect sizes are indicative only.

misof test

- **One dataset, one seed.** The script was set up for three replicates (v1–v3) × two seeds, but was stopped after v1 once the SPR arm's cost became clear. At its measured rate the remaining ten runs needed about 27 more hours.
- **Settings were trimmed for memory and time, equally in both arms.** The runs used a smaller starting-tree phase (`--ninit 10 --ntop 5`) and a 30-iteration patience (`-nstop 30`) instead of IQ-TREE's default 100. They also used `GTR+F` rather than the generating `GTR+F+I+G4`, which kept peak memory near 855 MB on a 7 GB machine but means the model is misspecified.
- **One SPR configuration.** The SPR arm used the settings recommended from the covid_500 bootstrap study. A smaller kick, or no `fullreopt` (which also removes the safe kernel), was not tried at full length.
- **The cost breakdown is partly measured on a subset.** The split into algorithm, `fullreopt` and safe-kernel costs comes from a 4,000-site subset at 1 thread. The finding that per-candidate cost grows with alignment length ($\approx 3\times$ at 4k sites, $9.1\times$ at 100k) is measured. The explanation offered, that SPR touches more partial-likelihood data per candidate and that data stops fitting in cache, is a hypothesis.
- **A saturated alignment says nothing about hard ones.** The result shows search is not the bottleneck on this simulated misof data. It does not show SPR is useless on alignments where search matters, such as covid_500.

Slack sweep

- **Unequal compute is the main caveat.** Every arm ran exactly 20 iterations, so the winners spent 6–15× the control's CPU. Without a stock run given the same CPU (roughly 150–300 iterations to match SA1), the sweep cannot separate "slack finds better trees" from "slack spends more compute".
- **Two seeds on one alignment.** Seed-to-seed swings reach 28 logL (SA5), as large as the effects themselves. The ranking of constant bound 1 against constant bound 5 flips between the seeds.
- **Three bugs were present during these runs** and were fixed on 11 September, after the sweep finished:
 - *Annealed slack stopped at 30% of its bound, not at 0 as documented.* The schedule counted kicks against a horizon that included the starting-tree iterations. Recorded final bounds: SA1 0.3 of 1, SA20 6 of 20, NSLACK5A 1.5 of 5. So every "annealed" arm here means "decayed to about a third".
 - *SA5 on seed 777 ran on a different schedule from SA5 on seed 1234.* It was interrupted when the laptop slept and resumed from its checkpoint, and the schedule's counter restarted with it, so its bound ended at

40% (2 of 5). That run is also the largest single gain in the sweep (+41.4), and its CPU figure may leave out a few minutes from before the interruption.

- *The accept-dist arms were handicapped.* Their temperature annealed from about 75% of T down to 5%, rather than from 100% to 0. More importantly, the NNI refinement had no best-tree restore, so each iteration handed back wherever its downhill walk happened to stop rather than the best tree it passed through. The stats files record `bestseen_restored=no` for every run, which here means "never checked". The accept-dist results are therefore pessimistic by an unknown amount.

All annealed-slack and accept-dist arms need re-running on the fixed build before being compared with anything new. The constant-bound arms (SC1, SC5, SC20, NSLACK5), the bare SPR kick and the control were unaffected.

- **The same arm cost 1.2–2× more on seed 1234 than on seed 777.** This is unexplained. It may come from seed-dependent numbers of refused kick moves, or from machine load.
- **Settings not varied:** kick strength (`--perturb 0.5` , i.e. 248 moves per kick on 500 taxa), SPR kick radius 6, the 8-redraw cap per kick slot, and bounds other than 1, 5 and 20.
- **A third seed exists but is not pooled here.** A separate run on another machine (seed 9102026, `-n 15`) agrees that all four matching arms beat stock: SPR-kick bound 5, +12.0 constant and +21.5 annealed; NNI-kick bound 5, +35.4 constant and +25.3 annealed. Annealing won for the SPR kick there and lost for the NNI kick. Its hardware and iteration count differ, so its CPU times are not comparable and it is kept separate.

Appendix · raw results and commands

Slack sweep, every run

Run	Final logL	CPU	Run	Final logL	CPU
CTRL_777	−84,382.068	230.2 s	CTRL_1234	−84,374.755	387.6 s
SPR0_777	−84,385.931	286.0 s	SPR0_1234	−84,380.171	565.4 s
SC1_777	−84,361.070	3,424.0 s	SC1_1234	−84,367.339	4,228.9 s
SC5_777	−84,368.908	1,602.0 s	SC5_1234	−84,359.543	3,216.6 s
SC20_777	−84,385.931	2,031.2 s	SC20_1234	−84,383.292	1,830.0 s
SA1_777	−84,354.090	5,016.5 s	SA1_1234	−84,344.975	4,480.8 s
SA5_777	−84,340.659	2,388.8 s	SA5_1234	−84,361.227	3,473.2 s
SA20_777	−84,385.931	1,784.8 s	SA20_1234	−84,383.292	2,175.4 s
NSLACK5_777	−84,374.565	1,791.4 s	NSLACK5_1234	−84,369.699	1,809.2 s
NSLACK5A_777	−84,367.577	1,848.0 s	NSLACK5A_1234	−84,355.399	1,825.8 s
AD05_777	−84,390.186	506.6 s	AD05_1234	−84,380.227	358.8 s
AD2_777	−84,395.895	358.8 s	AD2_1234	−84,374.166	538.4 s
AD2C_777	−84,395.715	357.2 s	AD2C_1234	−84,380.209	322.3 s

CPU is IQ-TREE's "Total CPU time used". Final logL is IQ-TREE's "BEST SCORE FOUND".

misof side benchmark (4,000-site subset, 1 thread, `-n 20`)

Configuration	Tree-search CPU	Candidates
NNI, plain kernel	28.6 s	18,168
NNI, <code>-safe</code>	39.8 s	18,168
SPR refine + kick, plain kernel	99.2 s	43,974
SPR refine + kick, <code>-safe</code>	110.1 s	43,974
SPR refine + kick, <code>-safe</code> + <code>fullreopt 100 20</code> (two runs)	176.5 s / 162.0 s	36,501

The two `fullreopt` runs did identical work (byte-identical trees), so their 9% difference is run-to-run timing noise on this machine.

Commands

```
misof, both arms:
iqtree3 -s ../run_misof_empirical_dna/concatenated_alignment.fasta -st DNA -m GTR+F -nt 2 \
  --ninit 10 --ntop 5 -nstop 30 -seed 777 [SPR arm: --spr-perturb "radius 6 weightprune" \
  --spr-refine "radius 10 fast quiet fullreopt 100 20 weightprune"]
```

slack sweep, every arm:


```
iqtree3 -s covid_data/covid_500.fasta -st DNA -m GTR+F -nt 2 --ninit 10 --ntop 5 -n 20 \  
-seed {777,1234} [arm flags from the table in section 2]
```

Where the files are

The raw logs, trees and `.searchstats.txt` files for both runs are in this session's scratchpad folder (`.../scratchpad/misof2/` and `.../scratchpad/slack_exp/`, with a text summary in `slack_exp/SLACK_SWEEP_SUMMARY.txt`). That folder is temporary; copy it somewhere permanent to keep the raw data.